CURRICULUM COMPANION : TECHNICAL LEARNING MODULES & FREE COURSES

# Week 1 Learning Materials : Foundations & Data Engineering

Core Concepts, Defensive Ingestion, Leak-Free Pipelines, 100% Free Courses & Documentation

| TARGET AUDIENCE | ESTIMATED READING TIME | COURSE COST | PRIMARY OBJECTIVE |
|---|---|---|---|
| ML Interns (Trial Phase) | 40-50 Minutes | 100% Free (Zero Paywalls) | Leak-Free Pipeline Discipline |

## 01 Virtual Environment Isolation & Atomic Git Hygiene

MODULE 1

Isolated environments prevent package conflicts between client projects. Pin exact versions in **`requirements.txt`** to ensure zero production drift. Commit code atomically following Conventional Commits format (`feat:`, `fix:`, `test:`, `docs:`). Never commit large raw data or virtual environment artifacts to version control.

```
# Virtualenv initialization and dependency management
$ python -m venv .venv && source .venv/bin/activate # On Windows: .\.venv\Scripts\Activate.ps1
$ pip install -r requirements.txt && pytest -v && ruff check .
```

## 02 Defensive Data Ingestion & Health Diagnostics

MODULE 2

Real-world telemetry data arrives with missing timestamps, corrupt rows, and unexpected encodings. Your loader must fail fast on invalid files and compute memory diagnostics before running training:

**DEFENSIVE INGESTION RULEBOOK**

**1. Defensive Type Casting:** Force string columns to strip whitespace; replace empty strings and "nan" tokens with explicit "UNKNOWN" categories.
**2. Skew-Aware Imputation:** Never use mean imputation on right-skewed operational metrics (such as session duration). Always use median values to resist extreme outliers.
**3. Memory Footprint:** Downcast 64-bit integers to 32-bit where allowable and convert low-cardinality strings to categorical dtypes to save 60-80% RAM.

## 03 Strict Data Leakage Safeguards & Stratification

MODULE 3

**Data Leakage** occurs when information from outside the training dataset is used to create or tune the model. A common mistake is fitting scalers or imputers on the whole dataset before splitting.

**THREE UNBREAKABLE RULES OF LEAK-FREE SPLITTING**

✓ **Split Before Transforming:** Partition data into train (70%), validation (15%), and test (15%) splits before calculating ANY mean, median, IQR, or scaler parameters.
✓ **Index Disjointness:** Formally verify with sets that `set(train_idx).isdisjoint(test_idx)` evaluates to True.
✓ **Stratified Sampling:** In imbalanced datasets (e.g. 15% fraud), random splitting can yield unrepresentative test subsets. Always stratify across the target label.

## 04 High-Performance Columnar Storage : Parquet vs CSV

MODULE 4

While CSV stores data row-by-row as uncompressed text, Apache Parquet uses columnar encoding with embedded schema metadata, snappy compression, and column pruning capabilities:

| FEATURE | CSV (COMMA-SEPARATED) | APACHE PARQUET (SNAPPY) |
|---|---|---|
| **Storage Format** | Row-oriented plaintext | Column-oriented binary |
| **Disk Footprint** | Large (e.g. 2.63 MB raw benchmark) | Compact (approx 0.7 - 0.9 MB with Snappy) |
| **Schema Preservation** | None; dtypes re-inferred on each load | Strict embedded schema & column dtypes |
| **Query Performance** | Must scan entire file sequentially | Reads only requested columns (column projection) |

## 05 Statistical Hypothesis Testing Frameworks

MODULE 5

Never rely solely on visual impressions from charts. Ground your findings in statistical hypothesis testing:

### TEST A Chi-Square Test of Independence ($\chi^2$)

Categorical vs Categorical

Evaluates whether two categorical variables (e.g. Device OS vs Fraud Flag) are statistically independent.

```
# Null Hypothesis (H0): Device OS and Abuse Rate are independent.
>>> chi2, p_val, dof, expected = scipy.stats.chi2_contingency(contingency_table)
# Decision: If p_val < 0.05, reject H0 (statistically significant association exists).
```

### TEST B Mann-Whitney U Test (Wilcoxon Rank-Sum)

Continuous vs Binary Label

Non-parametric test comparing the distributions of two independent groups when normality cannot be assumed (e.g. skewed session durations).

```
# Null Hypothesis (H0): Normal and Fraudulent sessions share identical distributions.
>>> stat, p_val = scipy.stats.mannwhitneyu(benign_durations, fraud_durations)
```

## 06 Unit Testing Data Pipelines & Code Quality

MODULE 6

- ☐ **Empty File Handling:** Verify `load_data` raises `ValueError` on 0-byte files.
- ☐ **Zero Data Leakage:** Assert train, val, and test index sets are completely disjoint.
- ☐ **Split Ratio Accuracy:** Verify sum of split lengths equals exactly total row count.
- ☐ **Documentation & Typing:** Every function has complete Google docstrings and type annotations.

**WEEKEND 1-ON-1 MENTOR PROBE PREPARATION**

During your weekend feedback session, your ML Lead will review your PR and ask: (1) "Why did you choose median over mean imputation for telemetry metrics?", (2) "Prove that your preprocessing pipeline introduces zero data leakage," and (3) "What did your Chi-Square contingency table reveal regarding device abuse?" Be prepared to walk through your code and charts directly.

## 07 Curated 100% Free Courses & Interactive Practice Labs

ZERO PAYWALLS

All courses listed below are **100% free** with no paywalls, subscriptions, or credit card requirements. They provide interactive coding environments and video walkthroughs directly aligned with Week 1 deliverables:

| COURSE & PROVIDER | PLATFORM & FORMAT | DURATION | CORE SKILLS & DIRECT APPLICATION |
|---|---|---|---|
| **Data Cleaning**<br>Kaggle Learn | Interactive Jupyter Labs | 4 Hours | Handling missing values, parsing dates, encoding cleanup, and inconsistent data entry resolution (Task 1.2). |
| **Pandas**<br>Kaggle Learn | Interactive Coding Labs | 4 Hours | Vectorized operations, multi-index sorting, grouping, summary metrics, and defensive data transformations (Task 1.1). |
| **Data Visualization**<br>Kaggle Learn | Interactive Jupyter Labs | 4 Hours | Seaborn distribution plots, correlation heatmaps, boxplots, and categorical bar charts for Task 1.3 insights. |
| **Missing Semester of CS**<br>MIT OpenCourseWare | Lectures + Exercises | Self-Paced | Shell scripting, virtual environments, Git version control, command-line data wrangling, and debugging hygiene. |
| **CS50P: Testing & Packaging**<br>Harvard University | Video + Sandbox Labs | 3-4 Hours | Writing automated unit tests with `pytest`, raising defensive exceptions, and modular package structure. |
| **StatQuest: Hypothesis Tests**<br>Josh Starmer (YouTube) | Illustrated Video Series | 2-3 Hours | Intuitive explanations of Chi-Square test of independence, p-values, Mann-Whitney U test, and IQR outlier boundaries. |

## 08 Official Documentation & Technical Cheat Sheets

PRIMARY REFERENCES

Bookmark these primary documentation hubs for immediate reference when designing your utilities and pipelines:

**PRIMARY DOCUMENTATION INDEX**

**1. Scikit-Learn Common Pitfalls:** `scikit-learn.org/stable/common_pitfalls.html` (Official guide on preventing data leakage and test-set contamination).
**2. Apache PyArrow Documentation:** `arrow.apache.org/docs/python/parquet.html` (Columnar format, Snappy compression, and zero-copy memory reads).
**3. Pandas Working with Missing Data:** `pandas.pydata.org/docs/user_guide/missing_data.html` (Nullable dtypes, boolean masks, and fillna patterns).
**4. Pytest Getting Started & Fixtures:** `docs.pytest.org/en/stable/getting-started.html` (Test organization, fixtures, and assertion reporting).
**5. Astral Ruff Documentation:** `docs.astral.sh/ruff/` (Modern high-speed Python linting, rule selection, and automated fixes).

## 09 Recommended 5-Day Study & Execution Plan

WORKFLOW SCHEDULE

- ☐ **Day 1 (Scaffold):** Watch MIT Missing Semester (Git/venv) & clone starter repository.
- ☐ **Day 2 (Utilities):** Review CS50P Pytest module & build `src/utils.py` with tests.
- ☐ **Day 3 (Cleaning):** Complete Kaggle Data Cleaning & generate `clean.parquet`.
- ☐ **Day 4 (EDA):** Watch StatQuest Chi-Square video & complete statistical tests in notebook.
- ☐ **Day 5 (Review):** Author Executive Insights Memo & push final PR before Friday 11:59 PM.
- ☐ **Weekend:** Review Mentor Probe Cheatsheet and attend scheduled 1-on-1 feedback session.